WIUT HACKATHON 2026 · COMPUTER VISION FINAL

OQIM

A traffic signal audit from one video

MODE: STARTUP

Team PariVision

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Live https://oqim-audit.vercel.app

Code https://github.com/baytoooo/oqim



A real frame of the Tashkent junction (organisers' footage, clip C3896), with detections from our elimination-stage system.

C3896 | cycle 75 s | 130 vehicles southbound

PROBLEM AND USER

Signal timing is rarely checked

~1,000
signalised junctions in Tashkent

125
connected to TsODD's adaptive system

545
planned to be connected in 2026

Presidential decree of 4 December 2025 on congestion. No measured results published so far. Sources: Anhor.uz, Yuz.uz

How a junction is checked today

- A manual study: people at the roadside count vehicles, then an engineer calculates.
- Most controllers are not connected, so their timing cannot be read centrally.
- US cities measure arrivals on green and split failures from controller logs (ATSPM). Here those logs are not available.

The user

TsODD's traffic engineer, and the contractors who do junction studies for it.

Who pays: the city, which already buys studies; road-safety and transport projects.

THE PRODUCT IN ONE PICTURE

One video in, a checked plan out

1

Video in

A clip of the junction. Draw the stop line, click the signal head.

2

Lights from pixels

The lit lamp inside the head's box. No controller access needed.

3

Measure

Crossings by side, waits before the line, arrivals on green, discharge, red-light runs, queue.

4

Replay check

The clip's own vehicles run through a queue model. It must give back the measured crossings.

5

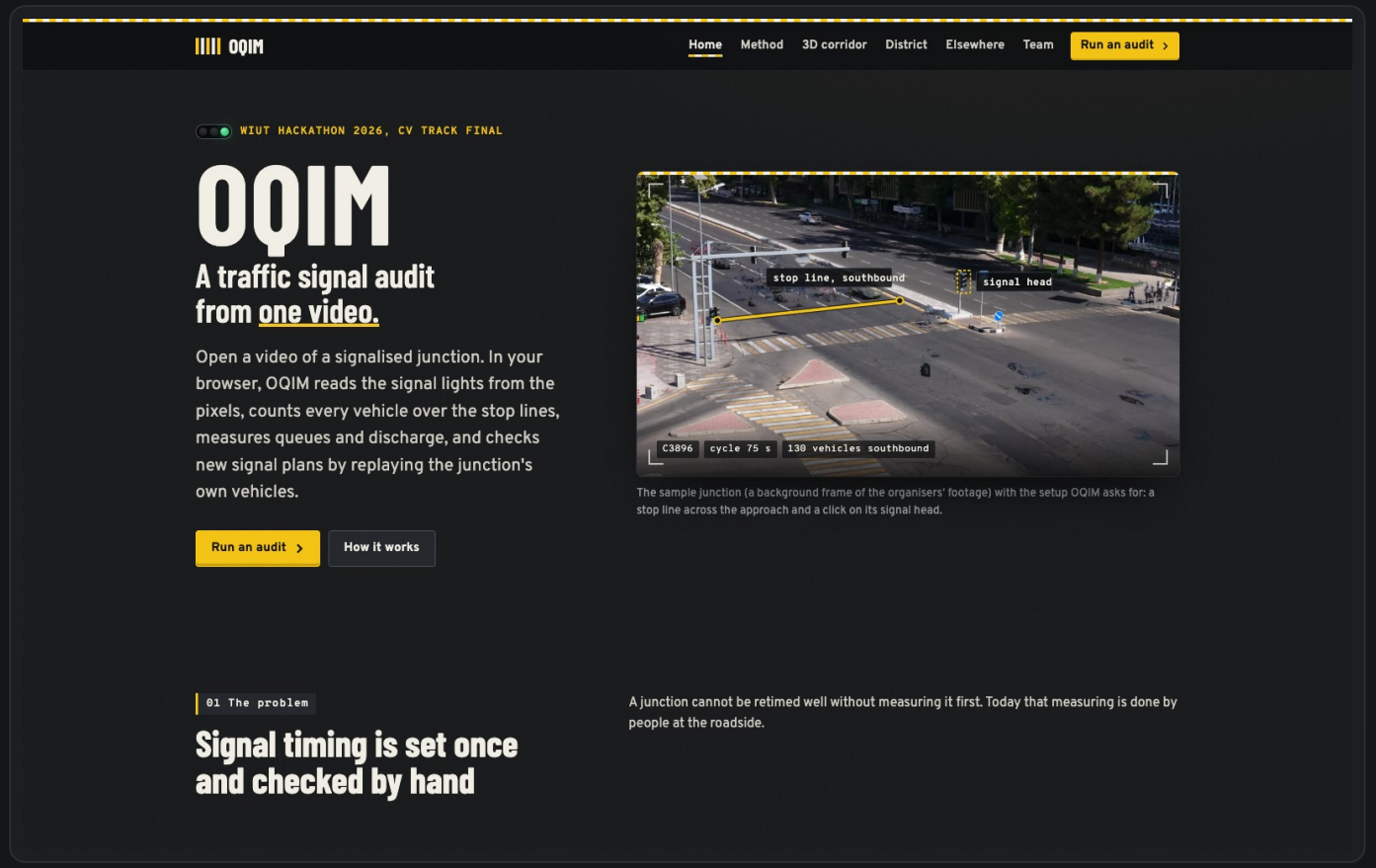
Report out

Six plans checked on the replay, traffic events, and a Purdue coordination diagram from video.

Runs on the user's device. The video is never uploaded. No number plates or faces are read.

LIVE DEMO

Live demo: five pages



- **/audit** — The jury's video: stop line, signal head, run.
- **/network** — Five real junctions of Tashkent, in 3D.
- **/city** — Kwun Tong: cameras and car park spaces.
- **/parking** — Car park spaces from camera frames.
- **/roads** — Road damage from phone photos, on a map.

https://oqim-audit.vercel.app Backup: the recorded run of C3896.

Everything runs in the browser

Browser tab on the user's laptop or phone

Video frames

5 frames per second, sent to a Web Worker. ffmpeg.wasm converts files the browser cannot decode.

Detect and track

YOLO26n (10 MB) by default, YOLO26s (38 MB) as an option. ONNX Runtime Web, WebGPU or WebAssembly. ByteTrack.

Audit core

TypeScript, no DOM. Lights, crossings, waits, discharge (busiest 10 s of each green), queue.

Replay

Point-queue model with the clip's own vehicles. Checked against the measured crossings.

Plans

Now, Webster (1958), searched fixed, actuated, max-pressure (Varaiya 2013), OQIM predictive.

5 fps

frames analysed per second of video

~65 s

to audit C3896 (5:40 of video) with YOLO26n on an Apple M5 laptop, WebGPU

0

server cost per audit. The site is static files.

Reused, declared: detector, ONNX worker, ByteTrack port from our elimination system.

Built for the final: audit core, replay, plans, corridor, district.

MEASURED ON OUR CLIPS

The replay gives back what we measured

Organisers' footage of one Tashkent junction, 18 minutes, southbound. Tracks: our elimination pipeline (YOLO26m, 1280 px, 10 fps).

Clip	Length	Cycle read	Vehicles	Model check: median departure error	Wait: model / measured
C3896	5:40	75 s	130	**0.50 s**	9.3 / 12.8 s
C3897	5:18	75 s	127	**0.29 s**	10.0 / 10.5 s
C3902	5:18	80 s	170	**0.55 s**	12.3 / 12.0 s
C3905	2:08	80 s	59	**4.54 s**	10.3 / 6.6 s

0.07 s

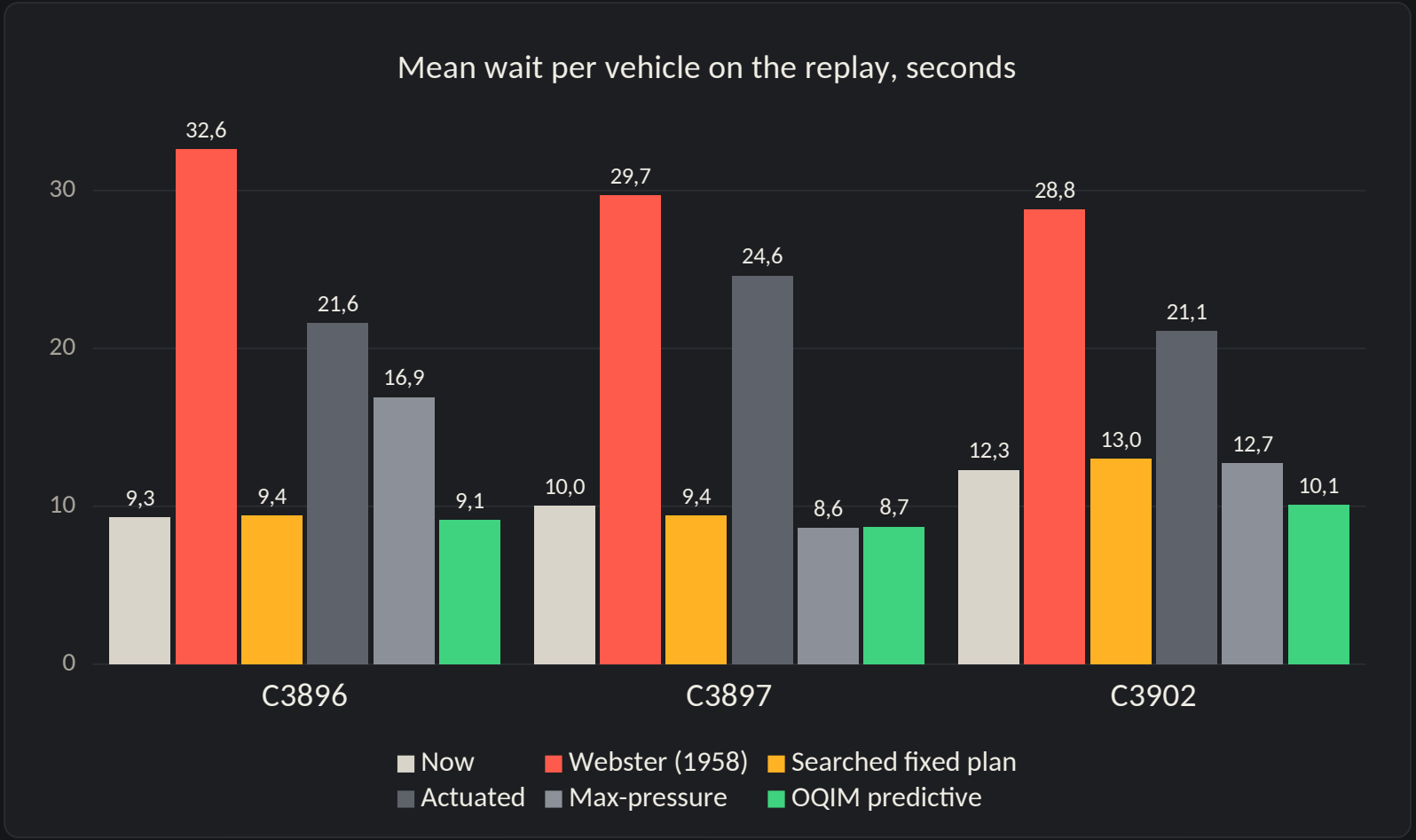
median departure error, browser run on C3896 today with YOLO26n (pipeline tracks: 0.50 s).

Known failure cases: short clips (under 2 cycles, see C3905), long queues where tracks break, a stop line drawn over two directions, free right turns counted as red crossings. Night and rain not yet measured on a test set of our own.

Same browser run: cycle 76 s, green 36 s, 127 vehicles over the line (Python pipeline: 130).

MEASURED ON OUR CLIPS

The textbook plan was worse than now



29.7 s vs 10.0 s

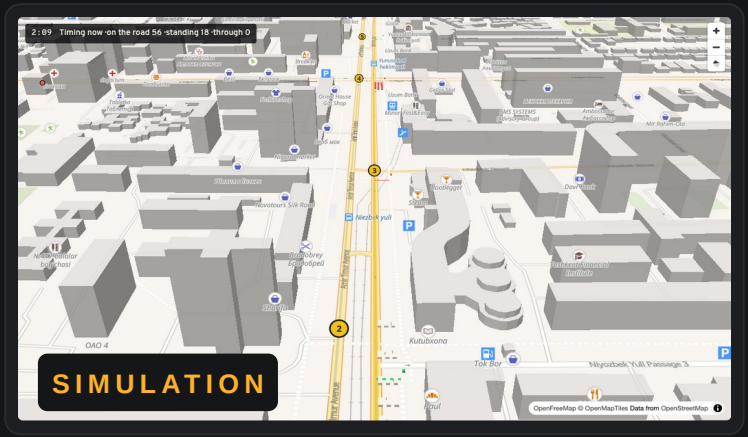
Webster (1958) against the timing now, C3897. Worse on all four clips.

8.7 s

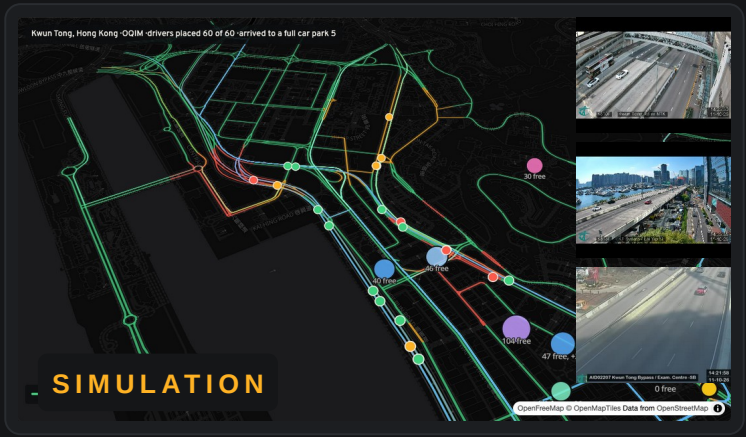
OQIM predictive, C3897. Now: 10.0 s. Searched fixed plan: 9.4 s.

So OQIM never recommends a plan before replaying it on the junction's own vehicles. Predictive assumes the next 30 s of arrivals are known.

A corridor, a district, a car park



Amir Temur Avenue, Tashkent: 1.46 km, 5 signals, 224 vehicles in 10 min. Green wave against timing now.



Kwun Tong, Hong Kong: 22 traffic cameras, about 20 car parks, 60 drivers to the MTR station.



/parking: free and taken spaces from ordinary car park camera frames, in the browser.

141 s → 24 s
mean delay at the signals

4.58 → 0.60
stops per vehicle

60 → 5
drivers arriving to a full car park

−18%
door to door

−41%
vehicle-km

97.4%
of spaces right, 240 held-out images (13,447 spaces)

99.1%
precision, taken

95.3%
recall, taken

Corridor and district are simulations. Tashkent: Intelligent Driver Model, one direction, only offsets change, real offsets unknown. Kwun Tong: drivers simulated; camera counts (YOLO26m, 960 px) and free spaces real, 13:45 Hong Kong time, 11 October. Parking: PKLot (CC BY 4.0, Almeida et al. 2015); sunny 97.0%, cloudy 99.0%, rainy 96.1%; no night frames in the dataset.

EVIDENCE

It already works elsewhere

Google Project Green Light

Up to 30%

fewer stops, up to 10% less emissions; about 70 junctions, 12 cities

Source: Seattle DOT

Surtrac, Pittsburgh

>25%

less travel time, idling down 40%

Source: Smart Cities Dive

SCOOT, UK

12%

less delay than up-to-date fixed plans, 20% than typical ones

Source: DfT TAL 7/99

Utah ATSPM

USD 108M

projected savings over 10 years, from the same measures as OQIM

Source: FHWA

Copenhagen green wave

17%

less travel time; stops from six to under one

Source: Centre for Public Impact

SFpark, San Francisco

50%

less cruising for parking, search time down 43%

Source: ACCESS Magazine

They can do it because they measure. Most junctions in Tashkent cannot be measured today. OQIM measures them from one video.

OUR EDGE

Why it is hard to copy

1

Lights read from pixels

No controller access. Built on our elimination work: signal reader, registration, rules, 98 labelled events.

2

The replay check

Plans are judged on the junction's own vehicles. The model must give back the measured crossings first.

3

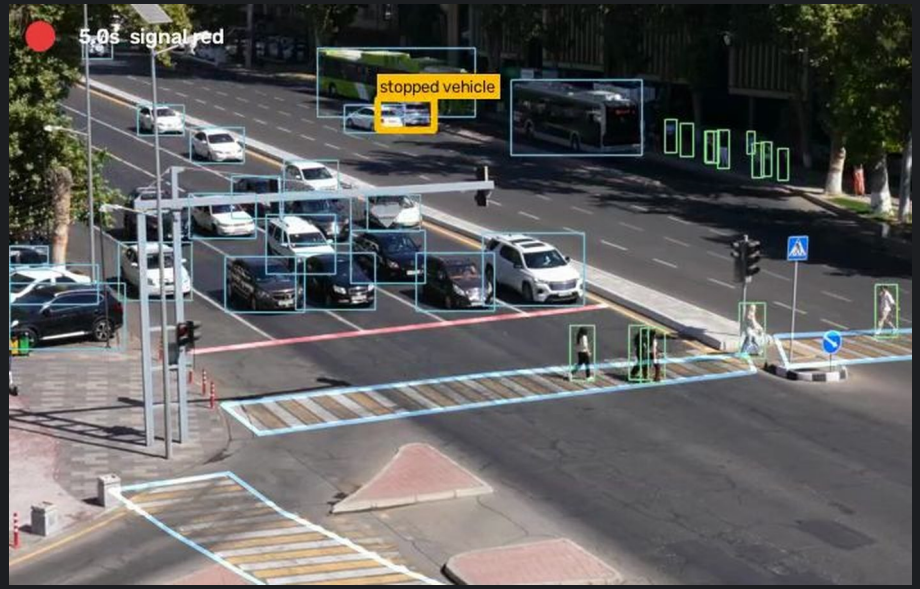
Runs in the browser

No server, no GPU bill. The video stays on the device, which suits Uzbekistan's personal data law; we store none.

4

Domain knowledge

Webster's method, saturation flow, ATSPM measures.



Our lamp reader on the organisers' footage, from pixels only.

98.8%

of frames of C3896: the browser's signal reading agrees with our elimination lamp reader.

Budget and ROI

Pilot with TsODD

3 months

20 junctions

People: 3 engineers part time. The main cost.

Hosting: static, free to USD 20 a month. No GPU server.

Data: videos the service already records or films for us. No licences to buy.

Per junction audit

0

compute cost. Inference runs on the user's device.

Our cost: an engineer's 30 minutes to film and check.

Price: below the cost of a manual study of the same junction. Price set in the pilot.

Value and payback

Value a day =

vehicle-hours saved per hour × hours a day at this traffic × (value of an hour + 0.8 l × fuel price)

Payback = price ÷ value a day

CO2: 2.31 kg per litre of fuel; idling at 0.8 litres an hour.

The engineer sets the inputs on the page. We state payback from the pilot inputs, not before.

AFTER THE HACKATHON

Next steps

1

Pilot

With TsODD: 20 junctions, their videos, their engineer marks the lines.

2

Test set

Our own, at night and in rain, with manual counts.

3

Network

Link neighbouring junctions (predictive needs the camera upstream), offsets for green waves.

4

Timing sheet

Export a sheet the engineer can enter into the controller.

Thank you. Questions?

Try it: https://oqim-audit.vercel.app Code: https://github.com/baytoooo/oqim